M4 Discussion: ML and DL — Fraud Detection in Retirement

Shortened draft by assignment section (~320 words)

1. Choose a Real-World AI Application

I am focusing on fraud detection in the retirement industry. Plan sponsors and recordkeepers use machine learning to protect 401(k), IRA, and pension accounts from unauthorized withdrawals, account takeovers, suspicious rollovers, and unusual beneficiary changes.

2. Analyze the Application

The problem is stopping fraud before money leaves the plan without blocking legitimate loans, hardship withdrawals, or rollovers. Training data may include contribution and withdrawal history, request type and amount, device or login location, advisor or call-center activity, and past cases labeled fraud or legitimate. Supervised machine learning fits best because the system learns from labeled examples and then classifies new requests. Lao (2018) explains that supervised learning uses labeled data and can solve classification problems—predicting categories such as fraud versus not fraud. Deep learning may help with very large, complex activity patterns. Reinforcement learning is less appropriate as the main method, since Lao describes it as reward-based trial and error rather than scoring new cases from labeled history.

3. Thinking About Model Performance

A more complex model can outperform a simpler one when fraud depends on combinations of signals, such as a large rollover plus a new device and a recent beneficiary change. If the model is too simple, it underfits and misses real fraud. If it is too complex, it can overfit old scam patterns and create false alarms on legitimate participant requests. Data quality matters because wrong labels, missing fields, or outdated behavior teach the model the wrong lessons.

4. Reflection

What I find most interesting is how supervised classification maps onto retirement operations: labeled past cases become predictions that protect people's savings. I was also surprised that "more complex" is not always better, because a false positive can temporarily block someone from their own retirement money.

References

1. Lao, R. (2018, January 22). A beginner's guide to machine learning. Medium. https://medium.com/@randylaosat/a-beginners-guide-to-machine-learning-dfadc19f6caf
2. [Add your course textbook / module reading if required.]

Most Applicable Sources

- Lao (2018) — best cite for supervised learning / classification vs. reinforcement learning

- Your textbook/module — best for model complexity, underfitting/overfitting, and data quality
- Optional: Fraud Detection Handbook (Lebichot et al.) — supervised fraud pipelines from labeled data